BACSE202 · Fall Semester 2026

PolyVote

A Multi-Tenant, Ranked-Choice Election Platform with a Tamper-Evident Audit Trail

Review 3 · Final Project Review · Guide: Dr. Katari Balakrishna



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What we submitted, what we built, what is new today

SUBMITTED

Reviews 1 and 2

Design, schema, ER and data-flow diagrams, literature; then a working React + FastAPI system, instant-runoff tally, hash-chained receipts and a live tamper test.

FEEDBACK

At Review 2

Confirm the voter with a second factor (OTP or face ID); finish MongoDB and hosting; show that the design is an improvement, not just a build.

NEW TODAY

This review

Atlas ledger, keyed chain, extended sweep, emailed one-time code, public link and QR, and 47 executable test scenarios with benchmarks.

47

test scenarios

Left open by choice: Oracle with PL/SQL and roles, the POSITION entity, and a true cloud platform. SQLite stays for this project.

Today: recap, new features, evidence, feature-by-feature comparison, privacy, limits, publication plan.

PICKING UP FROM REVIEW 2

Review 2 left a short list; two new features lead it

REVIEW 2 ENDED WITH

- About 75% of the scope built and demonstrated.
- Open: MongoDB Atlas, hosting, injected-ballot detection.
- Faculty feedback: confirm the voter with a second factor.
- Deferred on purpose: Oracle with PL/SQL, POSITION entity.

NEW: EMAIL OTP

A correct password is no longer enough.

A 6-digit code is emailed to the voter, valid 5 minutes, one use, 3 guesses.

Answers the Review 2 comment.

NEW: QR VOTING

Scan, sign in, vote.

The vote page shows a QR code for the public link, so any phone opens the ballot in seconds.

Works with the emailed code on the same phone.

RESPONSE TO FEEDBACK

Each Review 2 comment has a concrete answer

Comment at Review 2	What we did	Evidence
Confirm the voter: OTP or face ID	Emailed one-time code after the password; separate verification database; admin outbox	14 of 14 OTP tests pass
Finish MongoDB	Ledger now in MongoDB Atlas, unique-index concurrency control, file fallback, migration script	25 blocks migrated; live health endpoint
Host it online	Public HTTPS link through a tunnel; QR code on the voter page for phones	Cast a ballot from a phone
Prove it is better	47-scenario executable suite run on the Review 2 design and the final design	6 of 13 to 12 of 13 attacks caught
Weak spots we knew about	Injected ballots, forged receipts, rewritten chains, renamed candidates	Each has a test and a fix

Where we say "future work", it is because the design needs something we cannot add honestly in one semester, and we say what.

Small elections need ranked results and proof

THE PROBLEM

Small elections today

Clubs, councils and classes use form tools or one-table apps: one password per voter, results held by whoever has access, and no way to show that a vote was not edited afterwards.

THE GOAL

What PolyVote does

Run ranked-choice (instant-runoff) elections for many organisations, keep ballots anonymous in storage, and make later tampering detectable with checks anyone can rerun.

THE METHOD

Database first

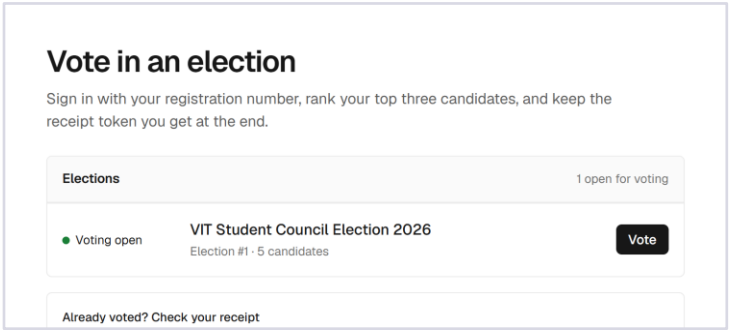
Rules live in the database (trigger, CHECK, UNIQUE, foreign keys); a second store holds a keyed hash-chain ledger; an executable suite attacks the design and reports what it catches.

47 executable test scenarios

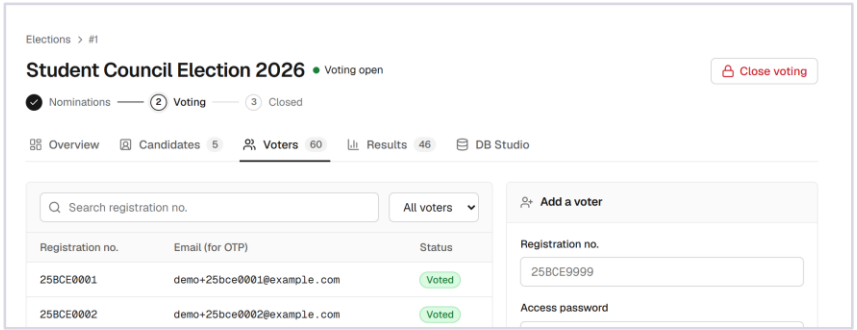
12 of 13 tampering attacks caught automatically

3 separate stores: election, verification, ledger

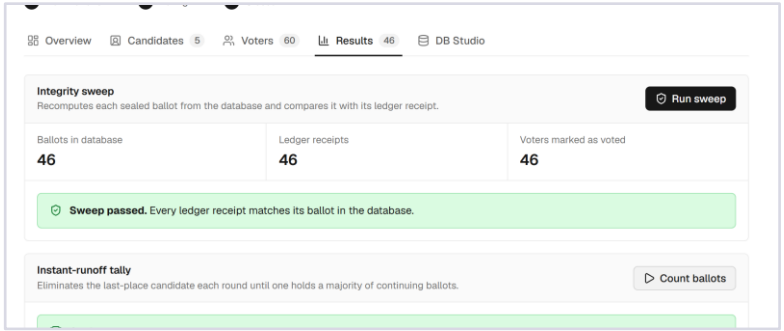
The whole system in four screens



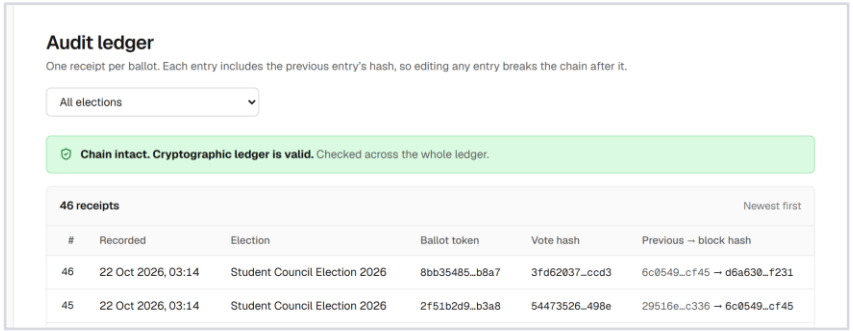
Voter site: open elections, receipt check, plain-language features



Admin: voter roster with the email used for the one-time code



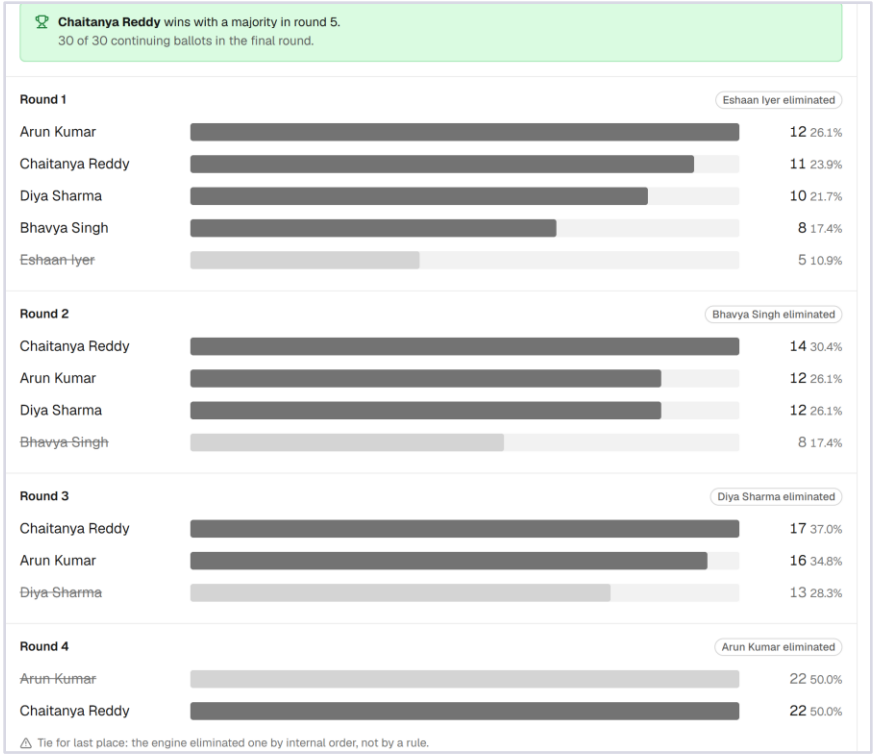
Admin: integrity sweep (46 / 46 / 46) and the instant-runoff tally



Admin: audit ledger, chain intact, 46 receipts

Instant-runoff counting, round by round

- **Rank:** each voter ranks up to three candidates.
- **Count:** first choices are counted; a majority of continuing ballots wins.
- **Eliminate:** otherwise the last-place candidate drops out and those ballots move to their next choice.
- **Repeat:** until someone holds a majority.
- **Shown in full:** every round, with counts and percentages, so anyone can follow the result.
- **Edge case, flagged:** round 4 here is a 22 to 22 tie; the engine eliminated one candidate by internal order and says so on screen. A published tie rule is future work.



THREAT MODEL

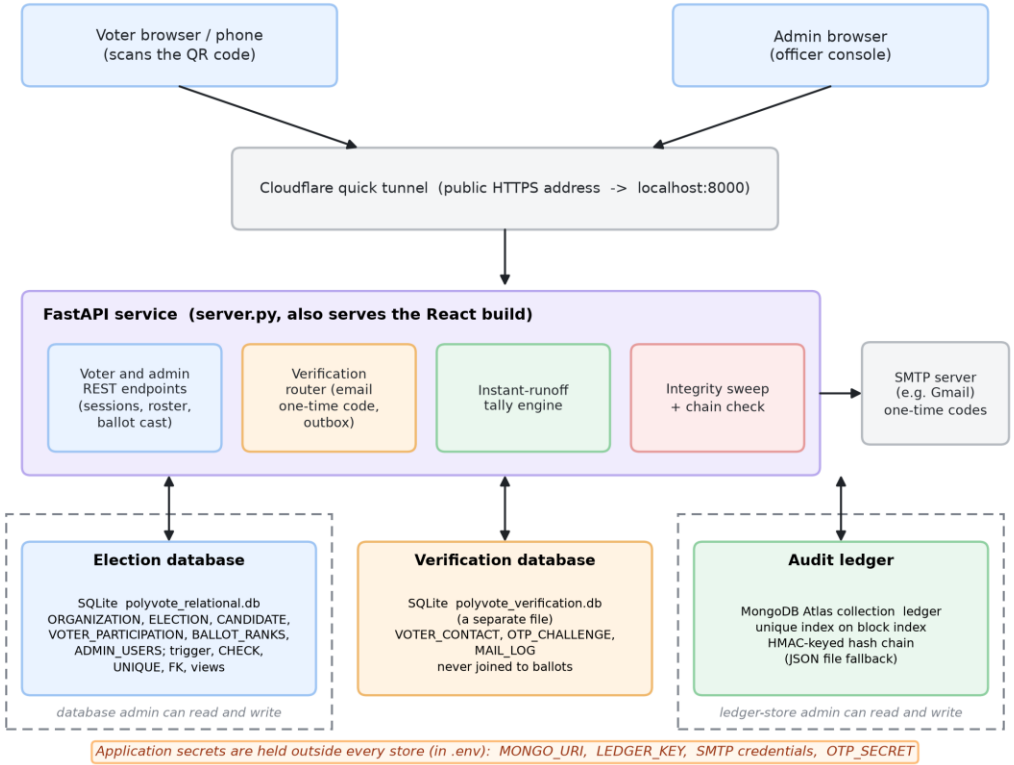
Who the attacker is, and who we do not stop

ID	Adversary	Capability	PolyVote response	Status
A1	Voter with a leaked or guessed password	Signs in as someone else	Emailed one-time code; one-vote trigger	Prevented
A2	Network observer	Reads or alters traffic	Out of scope beyond TLS at the tunnel	Not covered
A3	Rogue database administrator	Read/write on the election database only	Integrity sweep: modified, deleted, injected, mismatch	Detected
A4	Ledger-store administrator	Read/write on the ledger store only	HMAC hash chain; unique block index	Detected
A5	Insider with both stores, no secrets	Edits both stores to match each other	Keyed chain + sweep catch forged receipts, rewrites	Detected
A6	Insider with both stores and the ledger key	Forges valid blocks	Needs external anchoring (future work)	Not covered
A7	Roster injection through the admin path	Adds a ghost voter, votes as them through the API	Needs a committed roster (future work)	Not covered

Detected means found by an auditor running the checks, not blocked in real time.

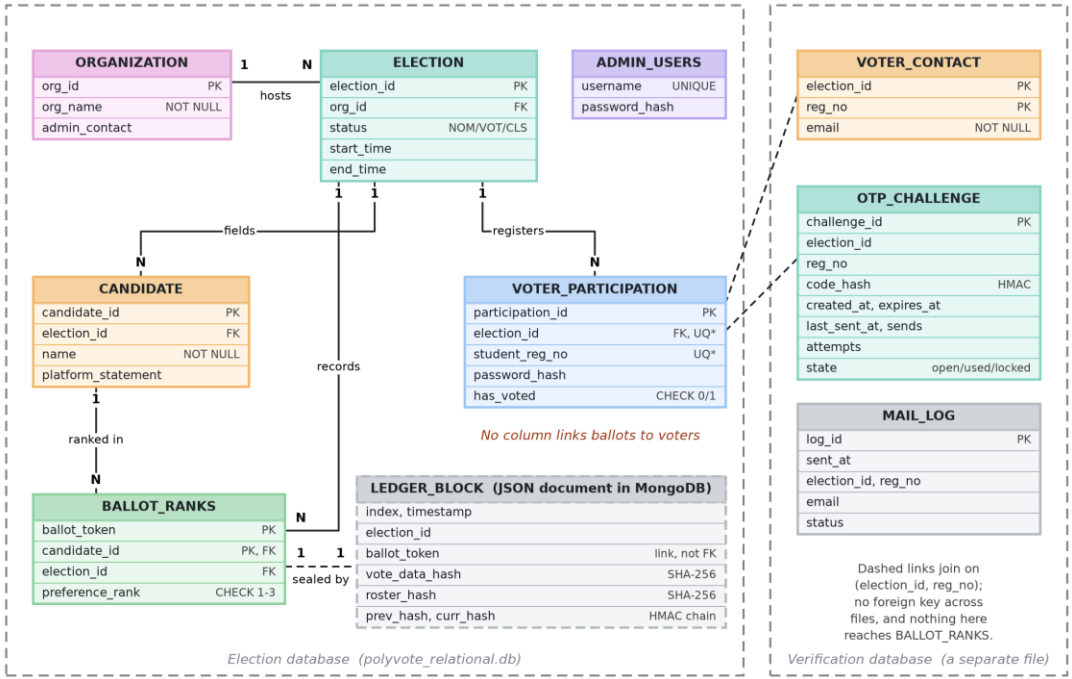
FINAL ARCHITECTURE

Three stores, separated by what each is trusted with



- **Election DB:** roster, candidates, anonymous ballots.
- **Verification DB:** emails and one-time codes, a separate file, never joined to ballots.
- **Ledger:** MongoDB Atlas, HMAC chain.
- **Secrets:** in .env, outside every store.
- **An administrator of one store cannot touch the others.**

Data model: election, ledger and verification stores

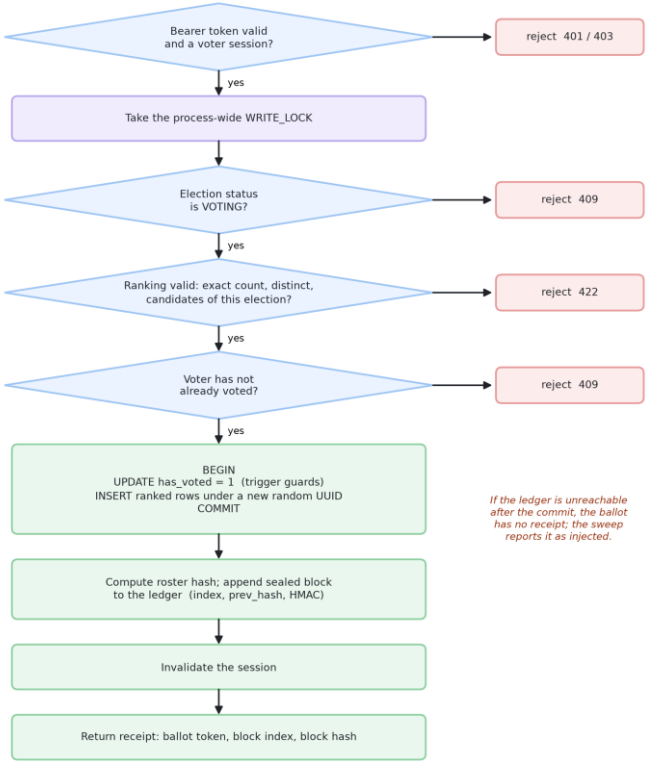


Solid line: foreign key. Dashed: logical link to another store (not an FK). PK primary key, FK foreign key, UQ unique, CHECK check rule.
* UNIQUE (reg_no, election_id). A trigger blocks marking a voter as voted twice.

- Shared multi-tenant schema, scoped by election.
- No column links a ballot to a voter.
- Candidate roster hash sealed in the ledger.
- Emails live in a different file; no key reaches ballots.
- Trigger, CHECK, UNIQUE, FK enforce the rules.

CASTING A BALLOT

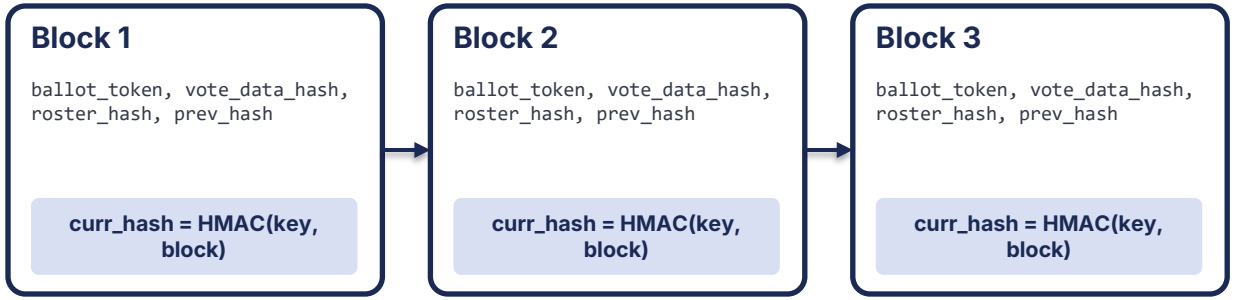
One serialised, transactional path touches every structure



- **Serialised:** a write lock plus the voted flag, checked inside the transaction.
- **Atomic:** flag and ballot rows commit together or not at all.
- **Anonymous:** the ballot is stored under a random token.
- **Sealed:** then a block with the ranks hash and roster hash joins the ledger.
- **Fails visibly:** if the ledger is down after commit, the sweep reports a ballot with no receipt.

KEYED AUDIT LEDGER

Keyed chain: rebuilding it needs the secret key



`prev_hash` links each block to the one before it.

- **Review 2:** curr_hash was plain SHA-256. An attacker who can write the store can edit a ballot and recompute every later hash.
- **Final:** curr_hash is HMAC-SHA-256 under LEDGER_KEY, held outside the database and the ledger store. Without the key the rebuilt chain fails verification.
- **Roster hash:** every block also seals the candidate list, so renaming candidates after voting is detected.
- **Backward compatible:** blocks written before the key existed still verify (LEDGER_LEGACY_UNTIL).
- **Limit:** whoever holds the key can still forge blocks; that needs an external witness (future work).

INTEGRITY SWEEP

Five rules compare the database with the ledger

Rule	Detects	How	New?
1 Modified	A ballot whose ranks changed	Recompute ranks hash; compare with the receipt	Review 2
2 Deleted	A sealed ballot missing from the database	Receipt exists with no ballot rows	Review 2
3 Injected	A ballot with no receipt	Ballot rows exist with no block	New
4 Roster changed	Candidate names altered after voting	Current roster hash differs from the sealed hash	New
5 Count mismatch	Flags, ballots, receipts disagree	Compare the three counts for the election	New

Plus the chain check: recompute every block's link and hash across the whole ledger. The chain check finds edits inside the ledger; the sweep finds edits to the database.

MONGODB ATLAS LEDGER

A real document store with its own concurrency control

- **Collection:** each block is one document in polyvote.ledger.
- **Unique index on block index:** two writers cannot both claim the next slot, so the chain cannot fork.
- **Append cost:** reads only the last block and inserts one document, so it does not grow with the chain.
- **Fallback:** no connection string, or Atlas unreachable, and it uses the local file; /api/health shows which.
- **Migration:** a script copies the old blocks and verifies the chain; it refuses to overwrite a non-empty collection.
- **Secrets:** the connection string lives in .env, never in the code or the repository.

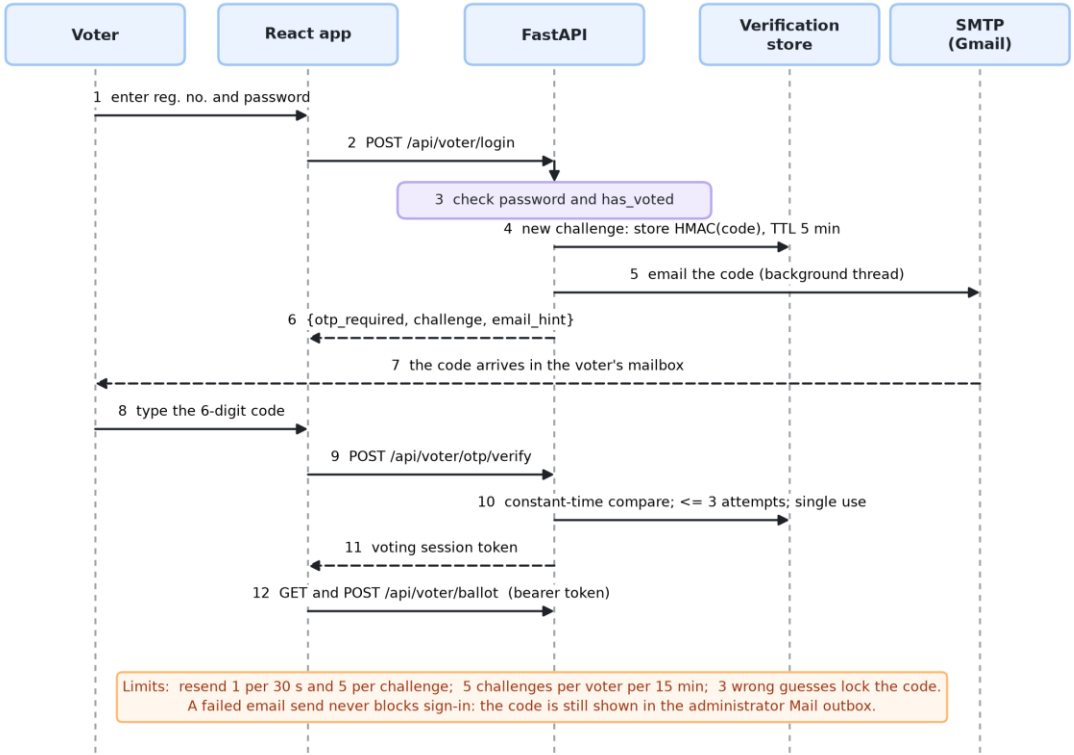
Live health check

```
{"backend":"mongodb",
 "detail":"mongodb
   (polyvote.ledger)",
 "sealing":"hmac-sha256"}
```

Editing any block in Atlas (for example a vote_data_hash) makes the Audit ledger page show "Data tampered at block N".

EMAIL ONE-TIME CODE

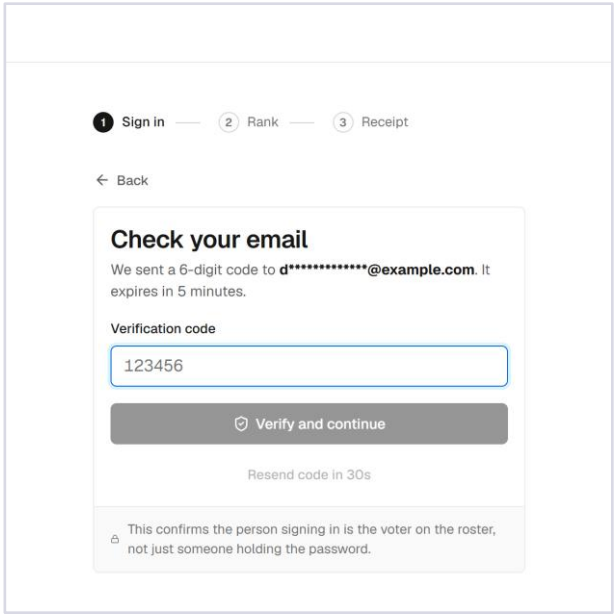
A correct password no longer opens a ballot on its own



EMAIL ONE-TIME CODE · DESIGN

Small choices that make it hold up

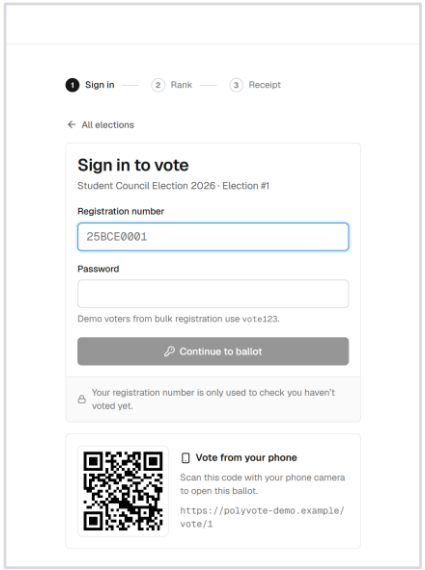
Choice	Value
Code	6 digits from a secure generator
Stored as	HMAC of challenge and code
Valid for	5 minutes, single use
Guesses	3 per code, then locked
Resend	1 per 30 s, 5 per challenge
Challenges	5 per voter per 15 min
Delivery	SMTP; failure never blocks sign-in
Where emails live	Separate database



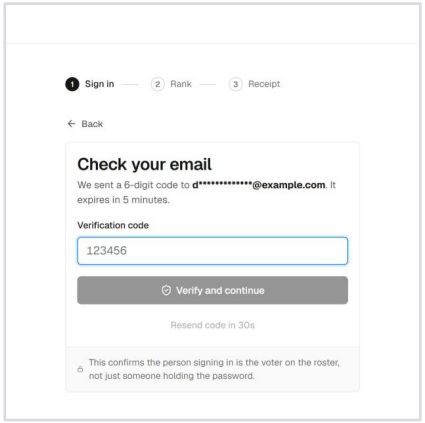
The voter sees only a masked address.

THE VOTER EXPERIENCE

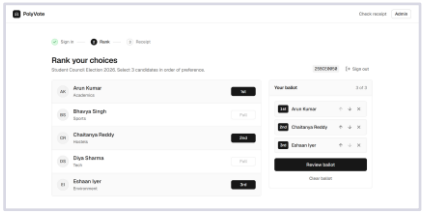
Sign in, confirm by code, rank, and keep a receipt



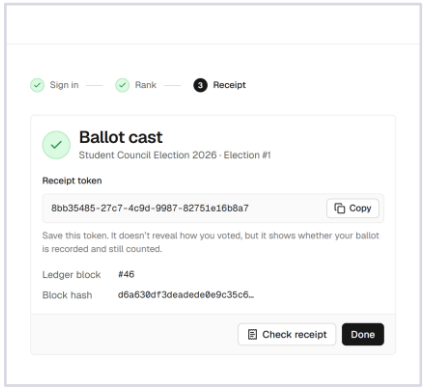
1 Sign in (QR for phones)



2 Enter the emailed code

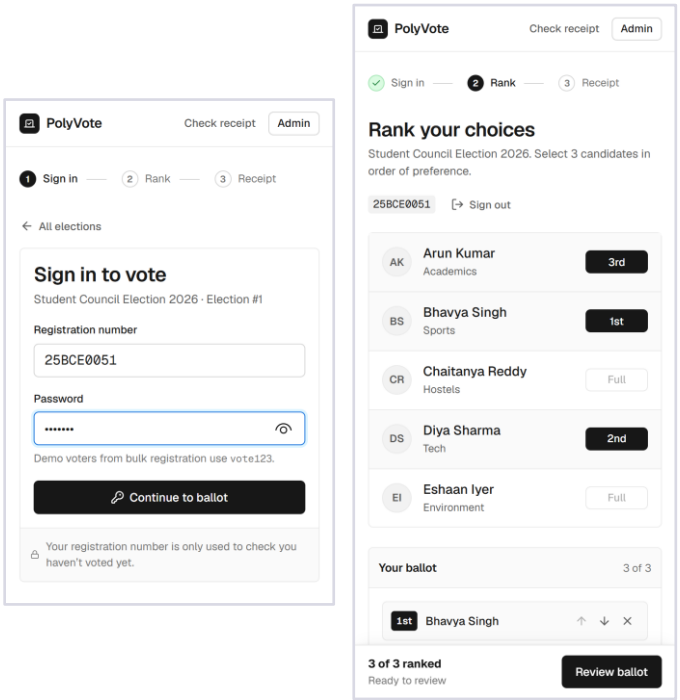


3 Rank three choices



4 Receipt token

Phone voting, a public link, and an honest account



- **How:** a Cloudflare quick tunnel gives a temporary public HTTPS address; the vote page shows a QR code for it.
- **Works from any network;** tested casting a real ballot from a phone.
- **What it is not:** cloud deployment. The laptop must stay on, and the address changes when the tunnel restarts.
- **Why we say so:** a true cloud platform with Oracle is the next step.
- **Safe by default:** secrets stay in .env; the repository holds only .env.example.

LIVE DEMO · SAMPLE POLYVOTE

The same system with email verification switched on

1

Admin

Voters have emails; Mail outbox open

2

Sign in

Password right, still no ballot

3

Code

Read it from the outbox or inbox

4

Vote

Rank, cast, get the receipt

5

Seal

Receipt check; new block in Atlas

6

Tamper

DB Studio edit; sweep names it

7

Ledger

Edit a block in Atlas; chain breaks

Safety net

Every code also appears in the outbox, so a slow email cannot stall the demo.

Separate copy: its own database, its own Atlas collection, port 8001; START_DEMO resets it in about 20 seconds.

HOW WE EVALUATED

47 executable scenarios, old design versus final

7	13	13	14	3
DB prevention	**API abuse**	**Tampering**	**One-time code**	**Benchmark sizes**
Statements the schema should reject	Wrong password, injection, replay, races	Edits by someone with storage access	Replay, expiry, limits, failure	100, 1,000, 10,000 ballots

- A throwaway copy of the app is driven through the real API; a "rogue administrator" edits the database and ledger files directly.
- Baseline: the Review 2 design, checked out and run unchanged. Attacker scripts never receive the ledger key.
- Outcome per attack: detected by the sweep or chain, visible only in counters, or not detected. Benchmarks are medians of three runs.

PREVENTION RESULTS

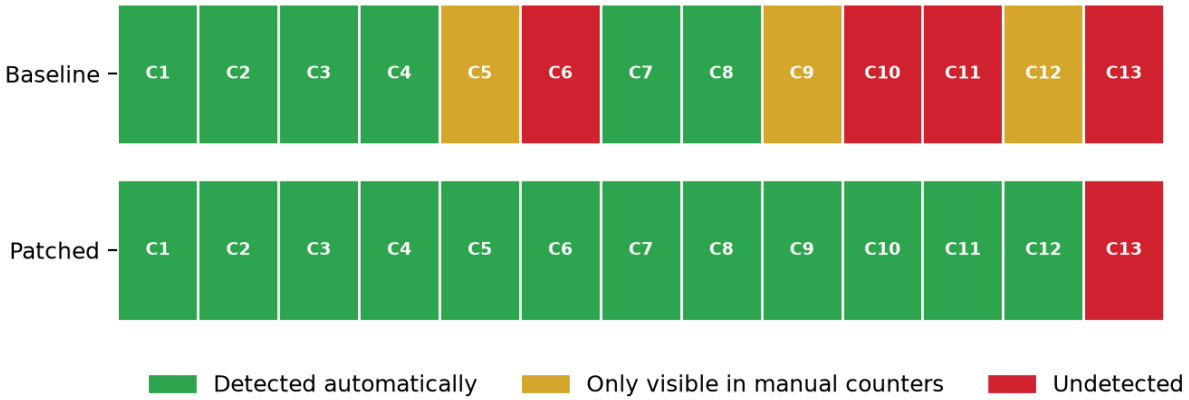
Prevention results: one gap taught us something

Attempt	Stopped by	Result
Second vote by the same voter	Trigger	Prevented
Rank 4 on a 1–3 ballot	CHECK	Prevented
Same candidate ranked twice	Composite key	Prevented
Duplicate roster entry	UNIQUE	Prevented
Wrong password, SQL injection, replayed session	API	Prevented
20 simultaneous casts by one voter	Lock + transaction	Exactly one accepted
Nonexistent candidate from a default console	FK (off by default in SQLite)	Not prevented

The gap: SQLite enforces foreign keys only on connections that enable it. The application does; a console with default settings does not. A server database removes this, which is part of the Oracle move.

TAMPERING RESULTS

Automatic detection rises from 6 of 13 to 12 of 13



6/13

Review 2 design: detected automatically

12/13

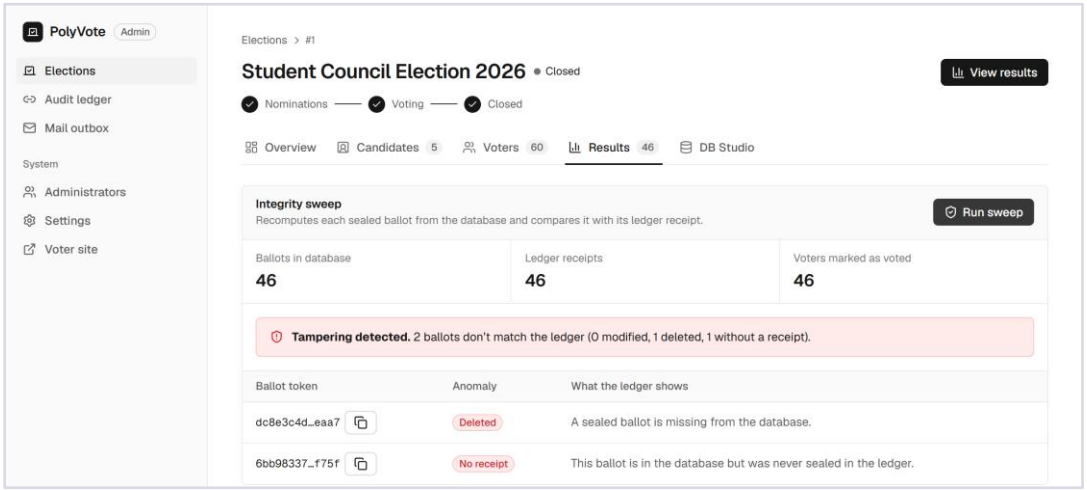
Final design: detected automatically

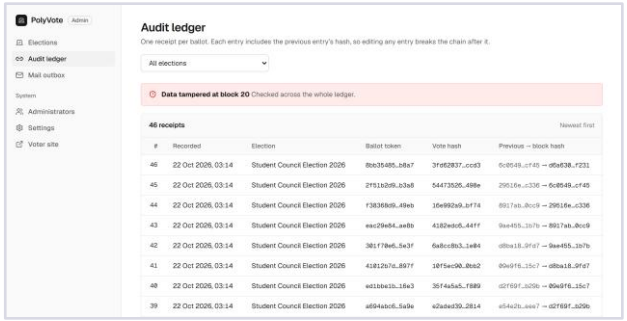
1

Still undetected: roster injection (C13)

SWEEP IN ACTION

A deletion and an injection hide in the counts





- **Left: counters read 46 / 46 / 46, yet the sweep reports 1 deleted and 1 without a receipt.**
- **Above: one edited block breaks the chain.**

WHY THE KEY MATTERS

Six scenarios the Review 2 design missed

ID	Scenario	Review 2	Final	What closed it
C5	Inject a ballot with no receipt	Counters only	Detected	Injected-ballot rule
C6	Ballot stuffing: inject, forge receipt, set a flag	Not detected	Detected	Keyed chain
C9	Truncate the last ledger block	Counters only	Detected	Reconciliation rule
C10	Alter a ballot and rebuild the whole chain	Not detected	Detected	Keyed chain
C11	Swap two candidates' names after voting	Not detected	Detected	Sealed roster hash
C12	Reset a voted flag, vote again	Counters only	Detected	Reconciliation rule

Key point: C6 and C10 defeat any chain that is only hashed, because the attacker recomputes the hashes. With the HMAC key they fail at the first block the attacker cannot compute.

ONE-TIME CODE TESTS

14 of 14 pass, and the suite caught a real defect on the way

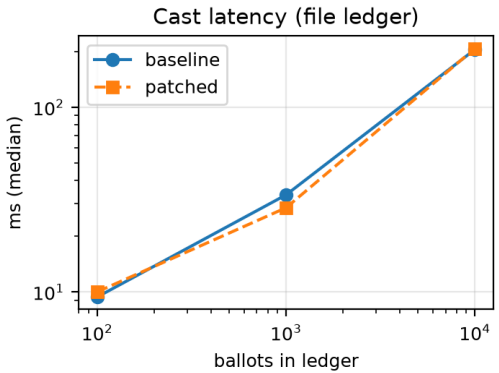
- Password alone gives no session; the challenge id is not a token.
- Right code opens the ballot; a used code cannot be replayed.
- Three wrong codes lock the challenge, even if the right code follows.
- Expired codes, rapid resends and a sixth code are refused.
- No email on file: clear refusal.
- Plaintext codes are absent from the database.
- A failed mail server does not block sign-in.
- No email column exists in the election database.

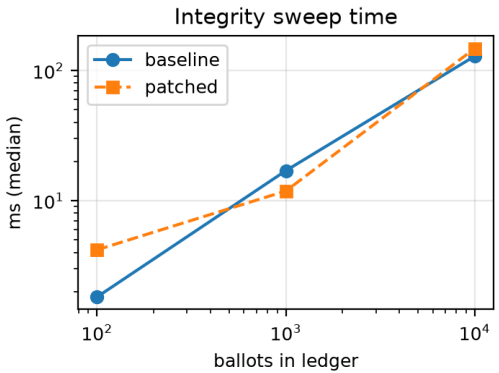
The defect

Test F4 first failed. The wrong-guess counter was saved inside a database context, and the request then raised an error. SQLite rolls back when an exception leaves the context, so every failed guess was recorded and then undone: unlimited guessing. The fix commits state before the error propagates. Review would not have found it; the test did.

PERFORMANCE

Checks are cheap; the file ledger is the one that grows





- Sweep at 10,000 ballots: 147 ms, about 15% slower than before.
- File ledger: a cast grows from 9 to 205 ms because each append rewrites the file.
- Atlas: about 147 ms at any size (network round trip).
- The file is faster when small; Atlas wins on flat growth, concurrency control and living off-machine.

Ballots	Cast ms (file)	Sweep ms	Atlas append ms
100	9	4	136
1,000	33	12	147
10,000	205	147	139

COMPARATIVE STUDY · SCOPE

We compared PolyVote with 33 systems in five groups

8 — **Verifiable and research**

Helios, Belenios, Civitas, ElectionGuard, Zeus, Verificatum, Vocdoni, Aragon

10 — **Web services and form tools**

CIVS, OpaVote, ElectionBuddy, Simply Voting, Election Runner, TapVoter, RankedVote, BetterVoting, Google and Microsoft Forms

2 — **Independent analyses**

Voatz (USENIX 2020) and Democracy Live (USENIX 2021)

2 — **Civic platforms**

Decidim and Consul Democracy

11 — **Open-source student projects**

FlexyVotes, VeriVote and nine other GitHub repositories

How

We read each project's own page, README or paper (23 web queries, 20 GitHub searches, 23 Oct 2026). A feature counts only if it is documented. "Not documented" is not proof of absence.

We did not run these systems. Every cell has a source URL in the evidence file; PolyVote's own cells rest on the tests in this deck.

COMPARATIVE STUDY · THE NINE FEATURES

Nine features, each with a way to test it

	Feature	How we prove PolyVote has it
F1	Ranked-choice counting with a round-by-round transcript	46-ballot election, 5 rounds shown, tie flagged
F2	Many organisations on one install	Tests B8 and B13
F3	Voting rules enforced inside the database	6 of 7 direct SQL attempts refused; 13 of 13 API abuses refused; 1 of 20 simultaneous casts accepted
F4	No link between ballot and voter; emails in a separate database	Automated check: no shared column, no email column in the election database
F5	Keyed audit ledger in a separate store, with a reconciling sweep	12 of 13 tampering attacks caught (plain chain: 6); identical in 3 runs
F6	Emailed one-time code as a second factor	14 of 14 tests, including replay, expiry and guess limits
F7	Scan-to-vote QR code on a phone	A real ballot cast from a phone through the public link
F8	Receipt to check a ballot is recorded and counted	Receipt check page; one ledger block per ballot
F9	Easy to run and host	One command locally; free hosted demo configured for Render

Coverage matrix: who documents which feature

System	F1	F2	F3	F4	F5	F6	F7	F8	F9
Helios	No	–	–	Part	Part	–	–	Yes	Part
Belenios	Part	Part	–	Part	Part	Part	–	Yes	Part
OpaVote	Yes	Part	–	–	–	Part	–	–	Part
ElectionBuddy	Part	Part	–	Part	Part	Part	Part	–	Part
TapVoter	Part	Part	–	Part	Part	Part	Part	Yes	Part
BetterVoting	Part	Yes	–	Part	Part	Part	Part	Yes	Yes
Google Forms	–	–	–	Part	–	No	–	–	Part
FlexyVotes	Yes	Yes	Yes	Part	Part	Yes	No	Yes	Part
VeriVote	–	–	Part	Part	Part	Part	–	Yes	Part
SecureVote	–	No	–	–	Part	Yes	–	–	Part
PolyVote (our tests)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Yes documented

Part partly

No absent

– not documented

F1 ranked counting · F2 many organisations · F3 database rules · F4 unlinked ballots · F5 keyed ledger and sweep · F6 email OTP · F7 QR voting · F8 receipt · F9 easy to run. Other rows: documentation we read. F5 is judged strictly (keyed chain, separate store, reconciling sweep, published attack results), so most systems score Part.

Ranked counting and many organisations: not unique

System	What its own page, README or paper documents	Match
OpaVote (web service)	Instant-runoff, STV and Condorcet with a step-by-step explanation of each count; free up to 25 voters; sub-groups only partly.	F1 Yes · F2 Part
FlexyVotes (GitHub)	Instant-runoff for one seat, STV for several, with per-round data stored; many organisations with scoped roles.	F1 Yes · F2 Yes
BetterVoting (GitHub)	Ranked-choice listed among voting methods; organisations and teams; round-by-round output not verified.	F1 Part · F2 Yes
Belenios	Condorcet-Schulze and STV are documented; instant-runoff and round transcripts are not.	F1 Part · F2 Part
Microsoft Forms	A ranking question exists, but no instant-runoff tally is documented.	F1 Part

PolyVote: counts instant-runoff and shows every round (46 ballots, 5 candidates, 5 rounds, tie flagged). Several organisations and elections ran in one install in tests B8 and B13.

Verdict: a baseline feature, not our claim. OpaVote and FlexyVotes also count ranked ballots, and OpaVote explains every count.

Database rules and unlinked ballots

System	What its own page, README or paper documents	Match
FlexyVotes (GitHub)	Unique indexes and append-only triggers that block even raw SQL; ballot rows carry no voter key. Voters and encrypted emails share one PostgreSQL database.	F3 Yes · F4 Part
VeriVote (GitHub)	Ballots stored without voter identity; triggers, CHECK and foreign-key rules are not documented.	F3 Part · F4 Part
CIVS (Cornell)	Voter emails are erased once the election begins; ballots are stored under a hash of the receipt.	F3 – · F4 Part
Helios, Belenios	Secrecy comes from encryption, not from a schema with no linking column; ballots sit on a public ballot box.	F3 – · F4 Part
Democracy Live (USENIX 2021)	The authors report voter identity, selections and a browser fingerprint sent to the vendor.	F4 No

PolyVote: 6 of 7 direct SQL attempts refused by the engine, 13 of 13 API abuses refused, exactly 1 of 20 simultaneous casts accepted. An automated check finds no column shared by ballots and voters; emails live in a separate file.

Verdict: FlexyVotes is stronger on database rules (append-only triggers). Our known gap is the one attempt not refused: SQLite foreign keys apply per connection.

Tamper-evident ledgers: where a keyed chain differs

System	What its own page, README or paper documents	Match
FlexyVotes (GitHub)	SHA-256 chain per organisation inside PostgreSQL, verified every 6 hours; Merkle bulletin board; 204 tests, no published attack table.	F5 Part
VeriVote (GitHub)	SHA-256 chain in MongoDB with an audit endpoint and a manual tamper demo; no HMAC key, no reconciliation.	F5 Part
SecureVote (GitHub)	Hash-linked ledger with hashed one-time codes; no ranking; a single instance by design.	F5 Part
BetterVoting (GitHub)	Audit log covers administrator actions (sharing a link, revealing a ballot link), not a chain of ballots.	F5 Part
Helios, Belenios	A public, cryptographically verifiable ballot box: a stronger guarantee by a different mechanism.	F5 Part
Voatz (USENIX 2020)	Claims a blockchain; the analysis found no public formal description and reported flaws.	F5 Part

PolyVote: an HMAC-keyed chain in a separate MongoDB store, a five-rule sweep and a published attack suite. 12 of 13 attacks caught against 6 of 13 for the plain-chain design, the same in 3 runs; C6 and C10 only fail with the key.

Verdict: better than a plain chain against someone who can write to the database. Not a match for cryptographic verification, and whoever holds the key can still forge.

Emailed codes exist elsewhere; so does QR sharing

System	What its own page, README or paper documents	Match
FlexyVotes (GitHub)	Keyed-hash one-time code with expiry, attempt limit and hourly rate limit; QR only for event ticket check-in.	F6 Yes · F7 No
SecureVote (GitHub)	Hashed one-time code with expiry and attempt limits.	F6 Yes
VeriVote (GitHub)	Email one-time code with rate limits.	F6 Part
BetterVoting	A unique emailed link per voter; docs mention no one-time code or 2FA; a QR code shares the election link.	F6 Part · F7 Part
TapVoter	Single-use key handed out by QR poster or email; no emailed second-factor code documented.	F6 Part · F7 Part
Belenios	Default voter login is an emailed short code at voting time; hashing, expiry and limits not documented.	F6 Part

PolyVote: password plus an emailed six-digit code (hashed, 5 minutes, single use, 3 guesses, resend limits). 14 of 14 tests pass, and one real defect (unlimited guesses) was found and fixed. A QR code on the vote page opens the ballot on any phone; a real ballot was cast from one.

Verdict: F6 alone is not unique: FlexyVotes and SecureVote have it too. What we did not find documented is a scan-to-vote QR flow together with an emailed code.

Receipts, cost and ease of running

System	What its own page, README or paper documents	Match
Helios, Belenios	Voters can check their ballot cryptographically, a stronger guarantee than a receipt. Helios needs PostgreSQL, RabbitMQ and about nine install steps.	F8 Yes · F9 Part
CIVS (Cornell)	Receipt per voter; free hosted service; MIT-licensed Perl code.	F8 Yes · F9 Yes
TapVoter	Receipt codes and a public transparency log; free tier up to 10,000 voters; closed source.	F8 Yes · F9 Part
BetterVoting	Receipt by email; AGPL licence; docker-compose in the repository; free hosted service.	F8 Yes · F9 Yes
FlexyVotes (GitHub)	Per-voter ballot tracker, Merkle proofs and an offline verifier; docker compose; no licence file found.	F8 Yes · F9 Part
OpaVote, ElectionBuddy	Free for 25 and 20 voters; closed source; no receipt documented.	F9 Part

PolyVote: a receipt proves a ballot is recorded and counted, not what it says, so it is weaker than Helios or Belenios. One command starts it (python -m uvicorn server:app); a free hosted demo is configured for Render; the source is public on GitHub, though without a licence file yet.

Verdict: easy to run, but BetterVoting and CIVS are also free and open. We do not claim an advantage here.

COMPARATIVE STUDY · CLOSEST PROJECTS

The three closest projects, and what each lacks

Closest project (source)	What it has	What we found it does not document
FlexyVotes, Django and PostgreSQL (github.com/deOpty/flexyvotes)	Many organisations; instant-runoff and STV; triggers that block raw SQL; ballots with no voter key; hash-chained audit log; hashed single-use email OTP	QR scan-to-vote; an HMAC-keyed chain; a separate document store and email database; a reconciling sweep; published attack results
VeriVote, Node, React and MongoDB (github.com/9022574361/Online_Voting_System)	SHA-256 chain in MongoDB with an audit endpoint; ballots without voter identity; email OTP with rate limits; receipt hash	Ranked counting; many organisations; database rules; HMAC key; reconciliation; automated attack suite
BetterVoting (github.com/Equal-Vote/bettervoting)	Ranked-choice voting; teams; QR sharing; receipts; free and open source (AGPL)	Database-enforced rules; a ledger of ballots (its audit log covers admin actions); one-time-code factor; reconciliation and attack tests

What we claim: every single feature exists somewhere, so PolyVote is not the first or the only one. We found no free small-election tool documenting all of them together. **What we do not claim:** to be more secure than Helios, Belenios or ElectionGuard. Limits: GitHub code search and non-English repositories were not covered.

COMPARATIVE STUDY · VERDICT

Best for small elections, not a Helios replacement

PolyVote IS A GOOD FIT WHEN

- A club, council or class needs ranked-choice results.
- Someone must be able to run it in an afternoon, free.
- You need proof that edits and deletions are caught.
- Voters should confirm themselves by email and vote from a phone.

CHOOSE HELIOS / BELENIOS WHEN

- The administrator itself cannot be trusted.
- Voters must verify their own ballot content.
- Coercion or vote selling is a real threat.
- A public cryptographic audit is required.

Our claim: for small elections, PolyVote adds database-level rules and tested tamper detection to an easy setup. It is not the strongest on verifiability.

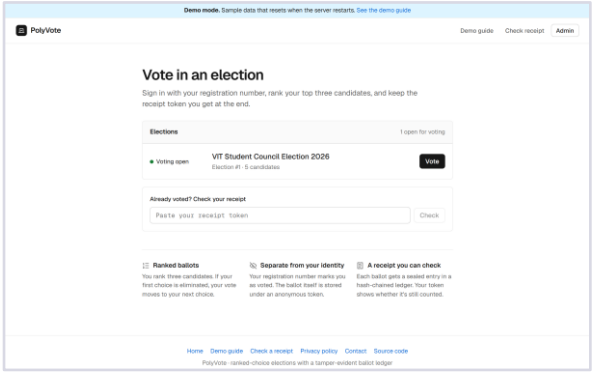
REVIEW 2 DESIGN VERSUS FINAL DESIGN

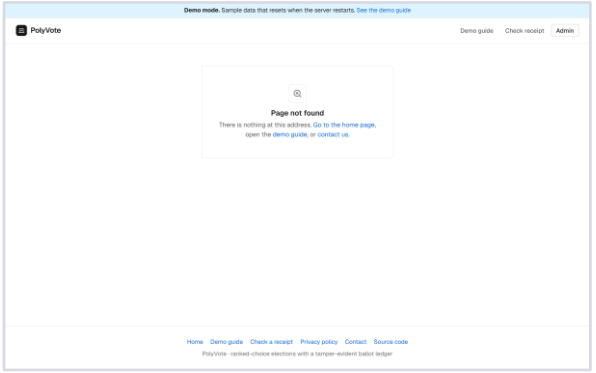
What improved, measured

Measure	Review 2	Final
Tampering detected automatically	6 of 13	12 of 13
Forged receipt or chain rewrite	Undetected	Detected (key)
Candidate renaming	Undetected	Detected (sealed roster)
Voter authentication	Password	Password + emailed code
Ledger storage	Local file	MongoDB Atlas
Ledger append at 10,000 ballots	205 ms (file, whole cast)	~147 ms (Atlas, flat)
Sweep at 10,000 ballots	129 ms	147 ms
Access	Local only	Public link + QR

Website launch checklist: 17 pass, 4 N/A, 0 fail

- PASS Server-side rendering (pre-rendered shell)
- N/A HTTPS and mobile viewport
- PASS robots.txt
- PASS sitemap.xml
- PASS Unique title tags
- PASS Meta descriptions
- PASS One H1 per page
- PASS Canonical tags
- PASS Open Graph images
- PASS Custom 404 page
- PASS 301 redirects
- PASS Alt text on images
- N/A WebP images
- PASS Lazy loading
- PASS Organization schema
- PASS Breadcrumbs
- PASS Internal links
- N/A Google Analytics
- N/A Search Console
- PASS Privacy policy
- PASS Working contact form





N/A: HTTPS (a local run; the host provides it), WebP (no raster images in the pages), Google Analytics and Search Console (need the owner's own IDs).

Privacy and ethics: what is stored, and where

Data	Where it lives	Stored as
Registration number	Election database	Roster entry
Password	Election database	SHA-256 hash only
Email address	Verification database (separate file)	Never stored with ballots
One-time code	Verification database	Keyed hash only; 5 min, one use
Your ballot	Election database	Random token; no voter column
Receipt block	Audit ledger	Token and hash of the ranks
Contact message	Verification database	Admins only

The ballot table and the voter table share no column, so the database cannot say who cast which ballot. A person with direct file access could read ballots, but not link them to people.

ETHICS AND LIMITS

- **Sample data:** the demo uses sample voters and resets on restart.
- **Officers see more:** the roster, emails and an outbox that lists codes for ten minutes. We kept it so a demo never stalls; a real election should switch it off.
- **Receipts:** prove a ballot is counted, not its content.
- **No tracking:** no tracking cookies; analytics is off unless the operator enables it.
- **Not for binding votes:** no independent review yet; passwords use plain SHA-256.

LIMITATIONS AND FUTURE WORK

What we do not claim, and what comes next

Limitation	Direction
Insider with key and both stores can forge (A6)	Publish the chain head to an external witness; managed secret store
Roster injection is not detected (A7)	Committed or signed roster, checked at sweep time
Not end-to-end verifiable; receipt shows recorded, not content	Cast-as-intended mechanism, or a verifiable protocol if needed
Passwords stored as plain SHA-256; default admin login	Salted, slow hashing; forced first-login change
Email is a weak second factor	Authenticator app (TOTP) or printed one-time slip
IRV ties broken arbitrarily; no statistical audit	Published tie rule; risk-limiting audit
SQLite foreign keys are per connection; single node	Oracle or PostgreSQL with PL/SQL, roles; cloud platform

GLOSSARY

Glossary in plain words

Term	In plain words
Hash	A fixed-length fingerprint of data. Change one character and the fingerprint changes completely.
HMAC	A hash made with a secret key, so only someone holding the key can produce a valid one.
Hash chain	Each record stores the previous record's hash, so editing one breaks every later link.
Ledger, block	The append-only list of receipts; one block for each ballot.
Instant-runoff	Rank candidates; last place drops out each round and its ballots move on, until someone has a majority.
OTP	One-time password: a short code that works once and expires.

Term	In plain words
Roster hash	A fingerprint of the candidate list, sealed into every block.
Receipt token	A random code that shows a ballot is recorded. It does not show the vote.
Trigger, CHECK, UNIQUE	Rules the database enforces itself, even against direct SQL.
Integrity sweep	Recomputes the election from the database and compares it with the ledger.
Polyglot persistence	Different kinds of database for different jobs: SQLite and MongoDB here.
Multi-tenant	Many organisations share one install, each with its own elections.

EVIDENCE

Evidence: where every claim can be checked

Claim in this deck	How it was produced	Where to look
Tampering caught: 6 of 13 (old), 12 of 13 (final)	Executable suite, 3 identical runs	experiments/results/summary.json, detection.png
Database refuses bad votes (6 of 7; 13 of 13; 1 of 20)	Direct SQL and API tests	experiments/run_experiments.py
14 of 14 one-time-code tests	Tests F1 to F14 on a throwaway copy	experiments/run_experiments.py
Sweep +15% at 10,000 ballots; Atlas append flat	Benchmarks, median of 3 runs	bench.png, mongo_bench.json
Website checklist 17 / 4 / 0	Automated check of 21 items	experiments/web_checklist.py, web_checklist.json
What 33 other systems offer	Their own pages, READMEs, papers; a URL for every cell	docs/review3/uniqueness_evidence.md
Closest projects and their gaps	FlexyVotes code and docs read directly	docs/review3/uniqueness_evidence.md

Search method: 23 web queries, 20 GitHub repository searches, 33 systems, on 23 October 2026. **Not covered:** GitHub code search, non-English and private repositories, commercial internals.

INDIA ONLY · NIRF TOP-100 HOSTS · OPEN TODAY

India only, NIRF top-100 hosts, cheapest first

#	Venue	Host university (NIRF 2025 rank)	Submit by	Student fee, INR	Held	Indexing
1	SCES-2027	MNNIT Allahabad, Prayagraj (Engineering 62)	15 Nov 2026*	5,900 incl. GST	12-14 Mar 2027	Claimed
2	IEEE COEP CON 2027	COEP Technological Univ., Pune (Engineering 90)	31 Mar 2027	6,000†	4-6 Jun 2027	Claimed
3	IEEE SANKALP 2027	ABV-IIITM, Gwalior (Engineering 96)	30 Nov 2026	7,000	9-11 Apr 2027	IEEE record #73342 shown
4	ICESICSC-2027	KIIT, Bhubaneswar (Overall 27)	30 Oct 2026	9,000	20-21 Feb 2027	Claimed
5	Confluence 2027	Amity University, Noida (Overall 37)	4 Dec 2026	9,440 incl. GST	21-22 Jan 2027	IEEE record #71526 shown
6	INDIACom-2027	BVICAM, Bharati Vidyapeeth, New Delhi (parent only, University 59)	7 Dec 2026	9,440 incl. GST	7-9 Apr 2027	Claimed
7	i-PACT 2027	VIT, Vellore (Overall 21)	31 May 2027	12,000	2-3 Sep 2027	Claimed
8	ICICC-2027	Shaheed Sukhdev College, Univ. of Delhi (parent only, Overall 15)	30 Oct 2026	13,000	5-6 Feb 2027	Claimed (LNNS "approved")

**Page marks the date tentative. †Fee read from a flattened table. "IEEE record shown" means a record number is on the official page, not checked on the IEEE site; nothing here is confirmed for IEEE Xplore. Also qualify, but off-topic or no published fee: ICSEAMS (Amrita), ICCIDS (SSN), PerCom work-in-progress (BITS Pilani Goa), NCC (IISc). Dropped, host not in the top 100: ICRCICN, IEM-ICDC, iConCISE, ICPCSN.*

CAN WE GET THE MONEY BACK?

Reimbursement: nothing is documented yet

WHAT WE FOUND

- **VIT:** the public Research Promotion Policy (2019) supports conference fees for faculty only, up to INR 20,000 a year in India. The student schemes (Raman Research Award, TRA fellowships) do not cover conference fees.
- **Organisers:** none publish a student waiver. Confluence, INDIACom and ICICC say fees are non-refundable.

WHAT TO DO BEFORE PAYING

- Ask the School Dean or Programme Chair, the Office of Academic Research and Student Welfare whether an undergraduate scheme exists, and get the answer in writing.
- Ask whether VIT credit needs IEEE Xplore indexing, before picking a venue.
- Until then, budget the fee as our own cost: about INR 5,900 to 9,440 for the cheaper venues.

The policy we read is from 2019; a newer or unpublished scheme may exist. We have not found one, and we are not assuming one.

RECOMMENDED SUBMISSION ORDER

One venue at a time: dual submission is not allowed

Step	Venue	Host university	Submit by	Why this order
1	IEEE SANKALP 2027	ABV-IIITM Gwalior	30 Nov 2026	IEEE record number shown; security and distributed-database tracks; INR 7,000
2	Confluence 2027	Amity University, Noida	4 Dec 2026	IEEE record number shown; INR 9,440; fast decision
3	ICESICSC-2027	KIIT, Bhubaneswar	30 Oct 2026	Closes in 7 days; only if the paper already fits their template
4	INDIACom-2027	BVICAM, Bharati Vidyapeeth	7 Dec 2026	Borderline host: only the parent university is ranked
5	i-PACT 2027	VIT, Vellore	31 May 2027	Our own campus, but a power-engineering conference

Before submitting: check the IEEE site for each record number, recheck the deadline and the NIRF edition (a 2026 list may be published), remove author names where double-blind, and follow each venue's AI-disclosure rule. Journals cost USD 2,000 to 3,000 in fees: only as a later extended version.

PROJECT CONCLUSION · REVIEW 3

Thank You

Questions and discussion

Source code and documents
github.com/Tharakeshtj/polyvote-election-platform

Run it: `python -m uvicorn server:app --port 8000`
Demo copy: Sample PolyVote, with email verification on
Reports: Review 3 report (44 pages) and IEEE-style paper